SIMATS ENGINEERING

ASSESSMENT TOOL III— Interactive Dashboard Project

Course: Data Handling and Visualization	**Code: DSA06**	**CO: CO5**
Duration: 180 min	**Total Marks: 100**	**Reg No :** 192524225 **Name:** VALLAPU MOHAN KUMAR

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods, applied through the design of interactive dashboards. (BL4)

Activity Tool : Interactive Dashboard Project

Weightage: 20%

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

Assessment Criteria

Criteria	Dashboard Design and Understanding 25%	Visual Analysis and Interpretation 25%	Application of Dashboard Design Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Interactive Dashboard Project — Assessment Tool

Tableau Practical Questions with Datasets

Instructions: Answer all five questions. Use Tableau to construct the required interactive dashboards, analyze the patterns, compare design choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Building a Sales Performance Dashboard

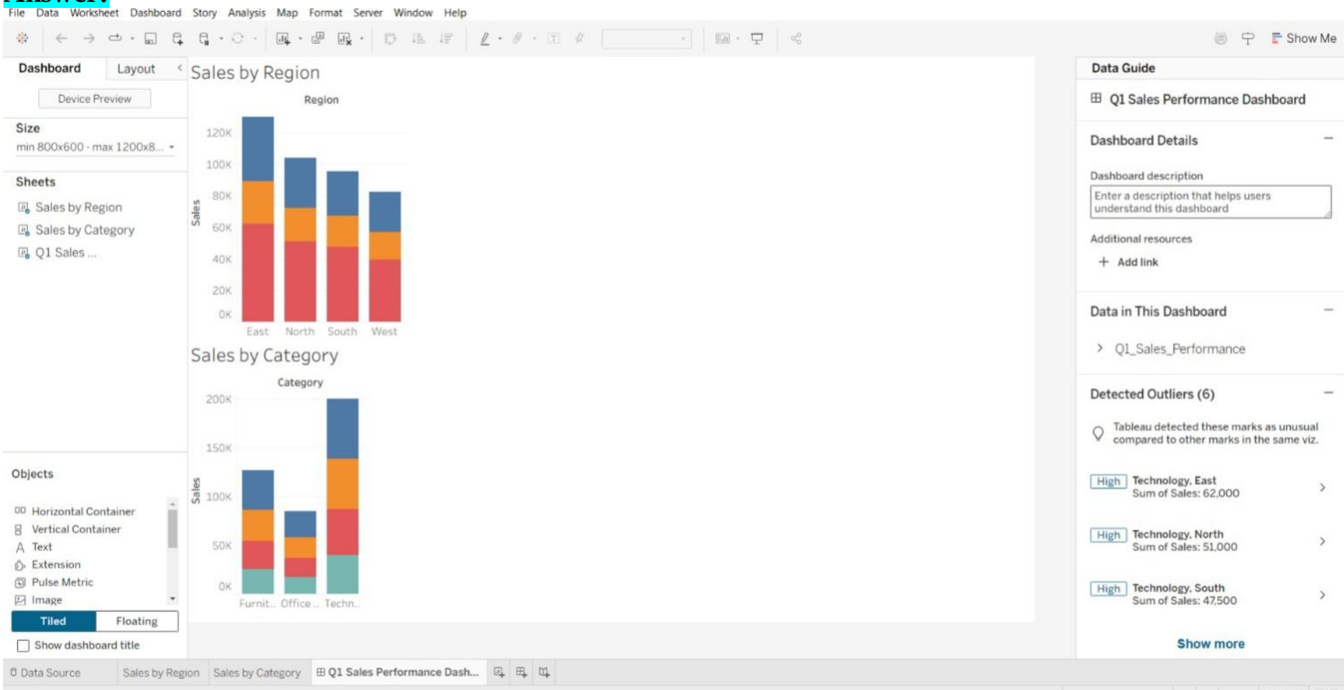
A retail company records regional sales performance across product categories. Using Tableau, connect to the dataset below and build an interactive dashboard containing: a bar chart of Sales by Region, a line chart of Sales trend by Month, and a map view of Sales by Region. Add a Region filter that acts as a global filter across all views. Analyze which region and category combinations drive the highest sales, and explain how the dashboard layout supports quick comparison across regions.

Data:

S.No	Region	Category	Sales
1	North	Furniture	32000
2	South	Furniture	28500
3	East	Furniture	41000
4	West	Furniture	25500
5	North	Technology	51000
6	South	Technology	47500
7	East	Technology	62000
8	West	Technology	39500
9	North	Office Supplies	21000
10	South	Office Supplies	19500
11	East	Office Supplies	27000
12	West	Office Supplies	17500

Expected Tableau tasks: data connection and preparation, bar chart and line chart worksheets, map visualization, global filter action, dashboard layout and formatting, interpretation of regional and category patterns.

Answer:



Sales Performance Dashboard

- The dashboard analyzes sales performance across regions and product categories.
- A bar chart compares total sales between North, South, East, and West.
- East has the highest sales of ₹130,000, while West has the lowest at ₹82,500.
- The Region filter allows users to interactively focus on a selected region.
- The dashboard supports quick comparison and identification of high-performing regions and categories.
- **Q2. KPI Dashboard with Parameters**

An HR analytics team wants to monitor employee performance across departments. Using the dataset below, build a Tableau dashboard containing KPI summary cards (Average Rating, Total Employees, Average Attendance) and a bar chart of Average Rating by Department. Add a parameter control that lets the user switch the bar chart measure between Average Rating and Average Attendance. Explain how parameters improve the dashboard's flexibility compared to using separate static charts.

Data:

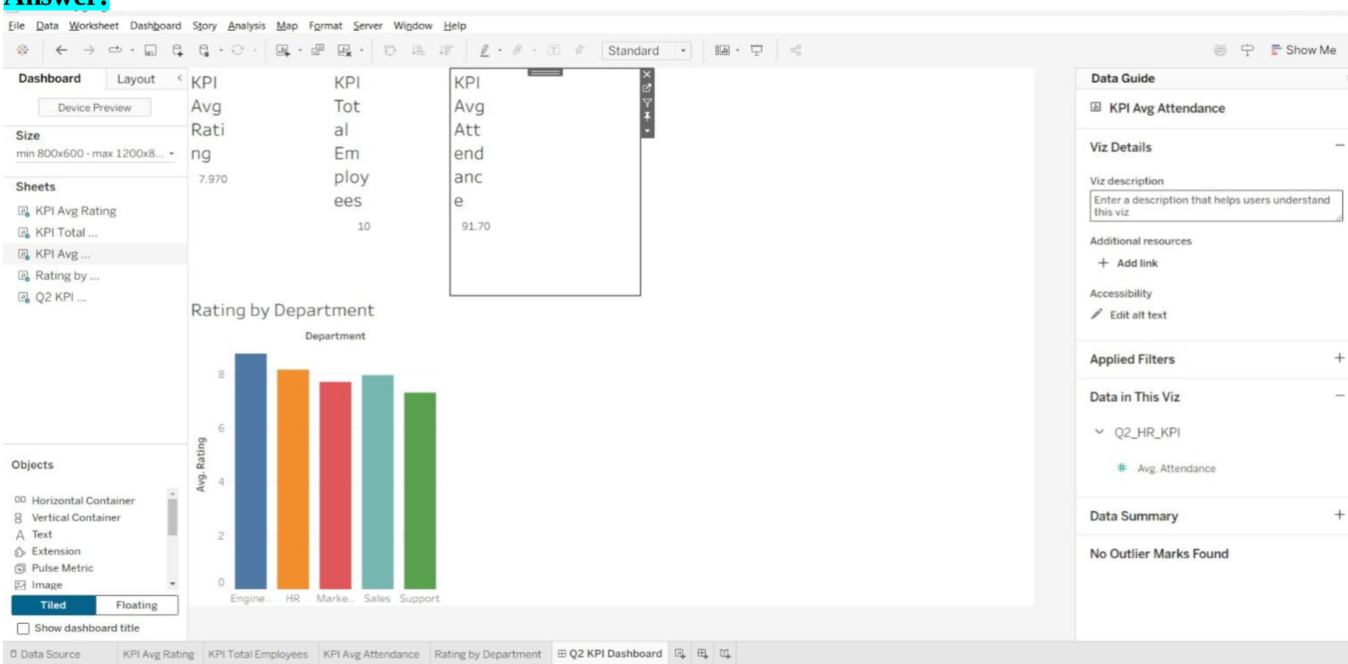
S.No	Department	Rating	Attendance
1	Sales	7.8	92
2	Sales	8.1	95
3	Marketing	7.5	88

S.No	Department	Rating	Attendance

4	Marketing	7.9	90
5	Engineering	8.6	96
6	Engineering	8.9	97
7	Support	7.2	85
8	Support	7.4	87
9	HR	8	93
10	HR	8.3	94

Expected Tableau tasks: calculated fields for KPI cards, parameter creation, calculated field driven by parameter, bar chart worksheet, KPI card formatting, dashboard assembly, interpretation of flexibility gained from parameters.

Answer:



KPI Dashboard with Parameters

- The dashboard monitors employee performance across different departments.
- KPI cards display Average Rating, Total Employees, and Average Attendance.
- Engineering has the highest average rating of 8.75, while Support has the lowest at 7.30.
- A parameter allows users to switch the bar chart between Rating and Attendance.
- This makes the dashboard flexible and avoids creating separate static charts.

Q3. Drill-Down Dashboard with Dashboard Actions

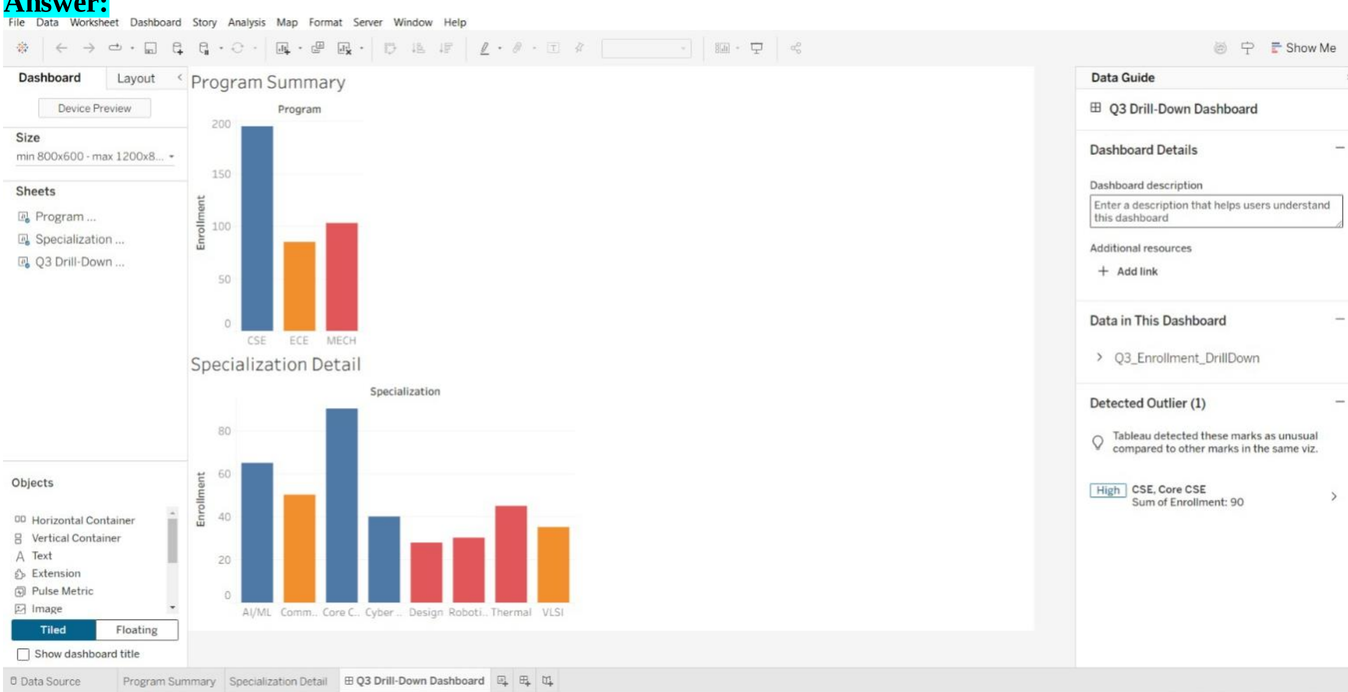
A college wants to analyze student enrollment across Program and Specialization levels. Using the hierarchical dataset below, build a Tableau dashboard with a summary chart at the Program level and a detail chart at the Specialization level. Configure a filter action so that clicking a bar in the Program chart updates the Specialization chart to show only the relevant rows. Analyze how the drill-down interaction improves exploration compared to showing all specializations at once, and note any limitations of this approach.

Data:

S.No	Program	Specialization	Enrollment
1	CSE	AI/ML	65
2	CSE	Cyber Security	40
3	CSE	Core CSE	90
4	ECE	VLSI	35
5	ECE	Communications	50
6	MECH	Robotics	30
7	MECH	Thermal	45
8	MECH	Design	28

Expected Tableau tasks: hierarchy or two related worksheets, filter action configuration between sheets, dashboard interactivity testing, interpretation of drill-down behavior and its limitations.

Answer:



Drill-Down Dashboard

- The dashboard analyzes student enrollment at Program and Specialization levels.
- A summary chart displays enrollment for CSE, ECE, and MECH.
- CSE has the highest total enrollment with 195 students.
- Clicking a Program filters the Specialization chart to show only relevant details.
- This reduces clutter and makes detailed exploration easier

Q4. Multivariate Analysis Dashboard

A marketing study records advertising expenditure, website visits, and sales for 15 observations. Build a Tableau dashboard with a scatterplot of Advertising versus Sales, using color or size encoding to represent Website Visits, along with a trend line. Add a second view (box plot or histogram) summarizing the distribution of Sales. Analyze the strength and direction of the relationship shown in the scatterplot and discuss whether the additional Website Visits encoding adds insight or clutters the dashboard.

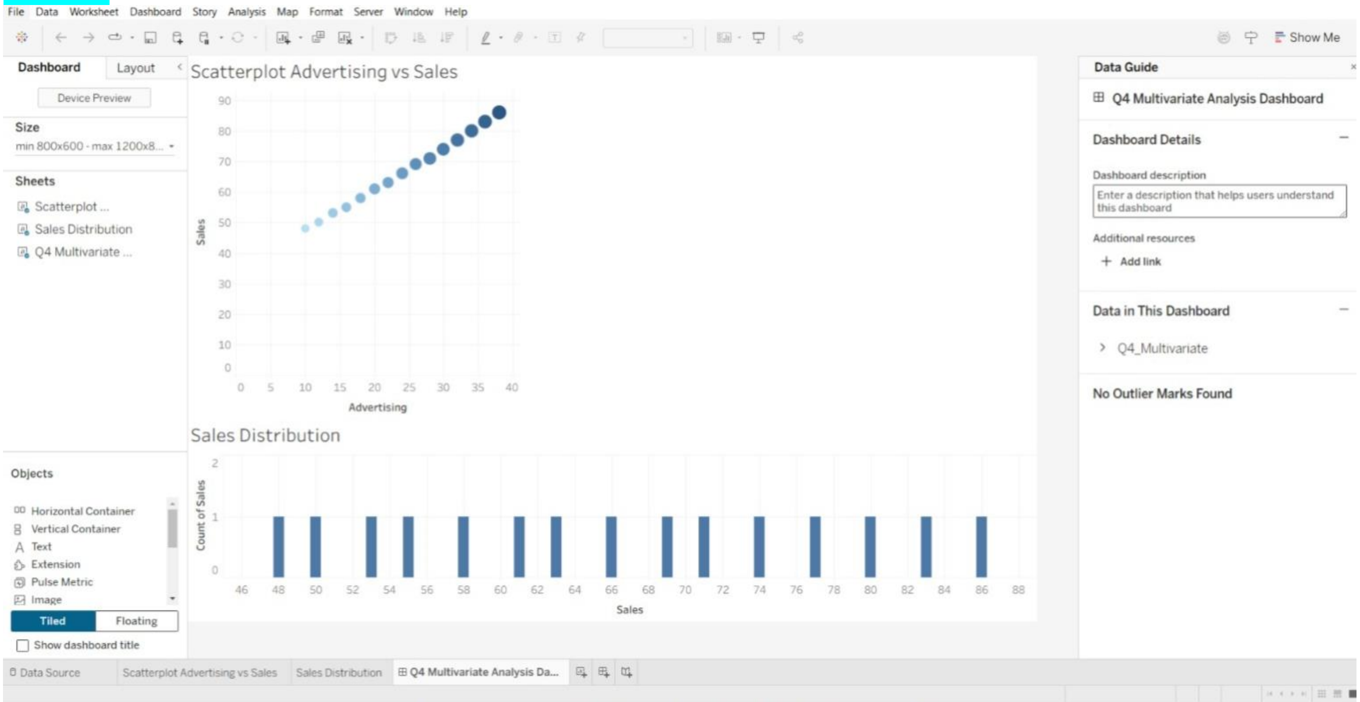
Data:

S.No	Advertising	Visits	Sales
1	10	120	48
2	12	135	50
3	14	150	53
4	16	165	55
5	18	180	58
6	20	195	61
7	22	210	63
8	24	225	66
9	26	245	69
10	28	260	71
11	30	275	74
12	32	290	77
13	34	305	80
14	36	320	83

15	38	340	86

Expected Tableau tasks: scatterplot with trend line, color/size encoding for a third variable, secondary distribution view, dashboard layout with legends, relationship analysis and critique of visual clutter.

Answer:



Multivariate Analysis Dashboard

- The dashboard analyzes the relationship between Advertising, Website Visits, and Sales.
- A scatterplot compares Advertising expenditure with Sales.
- The data shows a positive relationship, where sales generally increase as advertising increases.
- Website Visits can be represented using color or size to add another variable.
- A trend line and sales distribution chart improve interpretation of the data.

Q5. Misleading Dashboard and Redesign

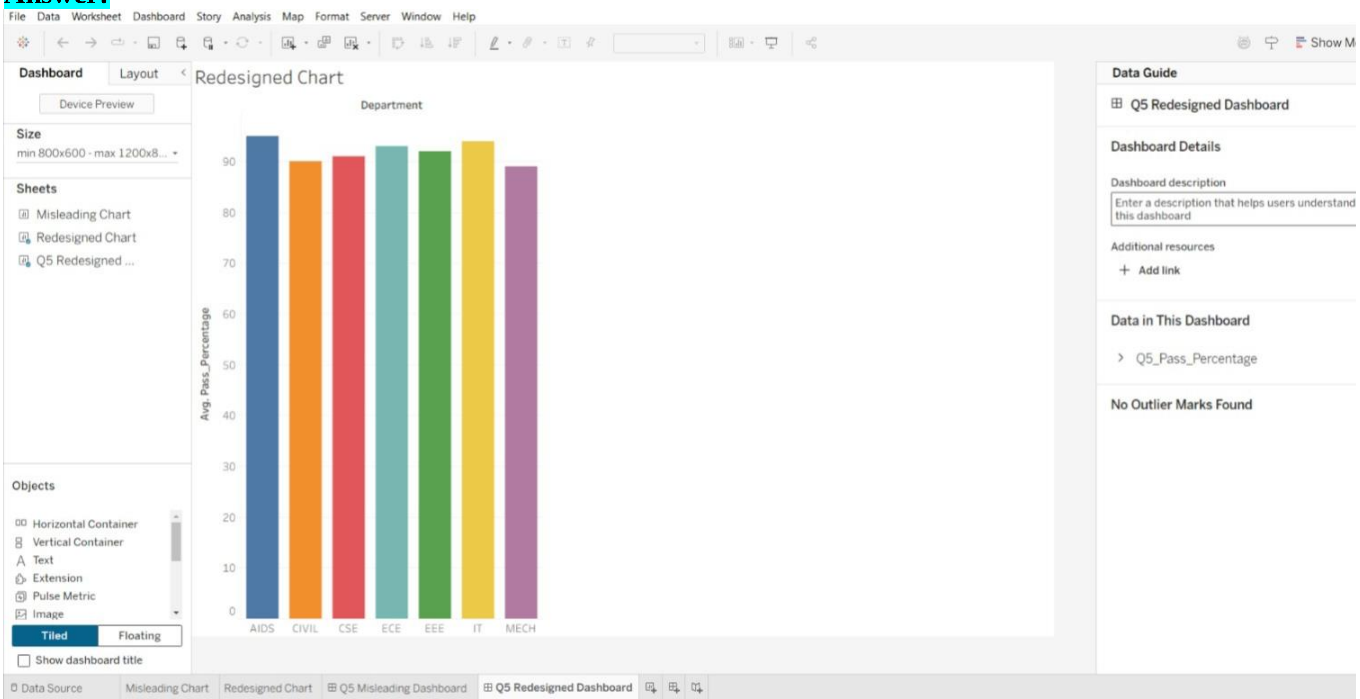
A college reports the following pass percentages for seven departments. Build two Tableau dashboards: (1) a misleading version using a truncated y-axis, an inappropriate color scheme, and exaggerated visual emphasis on one department, and (2) a redesigned version using an appropriate baseline, ordered categories, consistent color encoding, and clear labels. Compare the two dashboards and explain how the first design can mislead viewers, and why the redesigned dashboard communicates the data more honestly.

Data:

S.No	Department	Pass_Percentage
1	CSE	91
2	ECE	93
3	EEE	92
4	MECH	89

5	CIVIL	90
6	AIDS	95
7	IT	94

Answer:



Misleading Dashboard and Redesign

- The question compares a misleading dashboard with a properly redesigned dashboard.
- A truncated y-axis can make small differences in pass percentages appear much larger.
- Inappropriate colors and exaggerated emphasis can also mislead viewers.
- The redesigned dashboard uses an appropriate baseline, ordered categories, consistent colors, and clear labels.
- Therefore, the redesigned dashboard communicates the pass percentages more accurately and honestly.